

JKBOSE CLASS 10TH - MATHEMATICSSeries: 10th ARM(SZ) JKUT 2024 - Series A-3-Z | Code: 1003-Z | Total Questions: 40
Time: 3 Hours
Max Marks: 80 Marks**GENERAL INSTRUCTIONS:** Annual Regular Soft Zone Examination. Section A (1m), B (2m), C (3m), D (4m).**SECTION - A : OBJECTIVE & VERY SHORT QUESTIONS**

(1 Mark Each)

1. The number $0.10110111011110\dots$ is: [1 Mark]
(A) An even number
(B) A rational number
(C) An irrational number
(D) None of these
2. The product of zeroes of the polynomial $4x^2 + 8x$ is: [1 Mark]
(A) 2
(B) 0
(C) 4
(D) None of these
3. The pair of linear equations $x + 2y - 4 = 0$ and $2x + 4y - 12 = 0$ are: [1 Mark]
(A) Coincident
(B) Intersecting
(C) Parallel
(D) None of these
4. The relation $\sin 2A = 2 \sin A$ is true when angle A equals: [1 Mark]
(A) 0 deg
(B) 45 deg
(C) 30 deg
(D) None of these
5. The 11th term of the A.P. : -3, -1/2, 2, ... is: [1 Mark]
(A) 28
(B) 22
(C) -38
(D) None of these
6. The abscissa of any point lying on the Y-axis is: [1 Mark]
(A) 1
(B) 0
(C) -1
(D) None of these
7. H.C.F. of 26 and 91 is: [1 Mark]
(A) 26
(B) 13
(C) 14
(D) None of these
8. Getting a natural number greater than zero in a random experiment is an example of: [1 Mark]

- (A) An impossible event
- (B) A simple event
- (C) A sure event
- (D) None of these

9. The volume of a sphere of radius 'r' is: [1 Mark]

- (A) $(4/3) \pi \cdot r^2$
- (B) $(3/4) \pi \cdot r^3$
- (C) $(4/3) \pi \cdot r^3$
- (D) None of these

10. The discriminant of the quadratic equation $x^2 + 5\sqrt{5}x - 70 = 0$ is: [1 Mark]

- (A) 280
- (B) 405
- (C) 504
- (D) None of these

11. Prime factorization of 3825 is $3 \cdot 3 \cdot 5 \cdot 7 \cdot 17$. (State True or False) [1 Mark]

12. The sum of the first 1000 positive integers is: [1 Mark]

- (A) 500500
- (B) 500005
- (C) 100100
- (D) None of these

13. $1/2$ can be the probability of an event. (State True or False) [1 Mark]

14. All triangles are similar. (isosceles / equilateral) [1 Mark]

15. The number of tangents that can be drawn on a circle is [1 Mark]

16. If $a_n = (n - 1)(2 - n)$, find the fourth term a_4 . [1 Mark]

17. $x = 3, y = -2$ is a solution of equation $2x - 3y = 12$. (State True or False) [1 Mark]

18. The value of $\operatorname{cosec} A$ is always greater than or equal to 1. (State True or False) [1 Mark]

--- OR ---

Fill in the blank: $\sec^2 A = 1 + \dots\dots\dots$ (for $0^\circ \leq A \leq 90^\circ$).

19. Calculate the mean of the first 7 even natural numbers. [1 Mark]

20. Write the mathematical formula for the mean of grouped data. [1 Mark]

--- OR ---

Find the median of the observations: 6, 10, 14, 18, 22, 26, 30.

SECTION - B : SHORT ANSWER QUESTIONS**(2 Marks Each)**

21. Solve the pair of linear equations $3x + 4y = 10$ and $2x - 2y = 2$ by elimination method. [2 Marks]
22. Find the roots of the quadratic equation $2x^2 - x + 1/8 = 0$ by factorisation. [2 Marks]
23. Given $\sec \theta = 13/12$, calculate all other trigonometric ratios. [2 Marks]
24. Find the volume of a sphere of radius 3 cm. [2 Marks]

--- OR ---

Calculate the curved surface area of a cylinder of radius 2 cm and height 7 cm.

25. Find the values of y for which the distance between the points $P(2, -3)$ and $Q(10, y)$ is 10 units. [2 Marks]
- Check whether $(5, -2)$, $(6, 4)$ and $(7, -2)$ are the vertices of an isosceles triangle.
26. Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $1/3$, respectively. [2 Marks]

--- OR ---**SECTION - C : SHORT ANSWER QUESTIONS****(3 Marks Each)**

27. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts. [3 Marks]
28. Find the area of a quadrant of a circle whose circumference is 22 cm. [3 Marks]
29. Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. [3 Marks]
- Two tangents TP and TQ are drawn to a circle with centre O from an external point T . Prove that $\angle PTQ = 2 \times \angle OPQ$.
30. E is a point on the side AD produced of a parallelogram $ABCD$ and BE intersects CD at F . Show that $\triangle ABE \sim \triangle CFB$. [3 Marks]
31. The diagonals of a quadrilateral $ABCD$ intersect each other at point O such that $AO / BO = CO / DO$. Show that $ABCD$ is a trapezium. [3 Marks]
32. Prove that $6 + \sqrt{2}$ is an irrational number. [3 Marks]
33. An A.P. consists of 50 terms of which the 3rd term is 12 and the last term is 106. Find the 29th term. [3 Marks]

--- OR ---

Find the sum of the first 15 multiples of 8.

34. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting: (i) A face card, (ii) A spade.

[3 Marks]

SECTION - D : LONG ANSWER QUESTIONS

(4 Marks Each)

35. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the speed of the train.

[4 Marks]

--- OR ---

Find the value of K so that the quadratic equation $Kx(x - 2) + 6 = 0$ has two equal real roots.

36. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm³ of iron has approximately 8 g mass. (Take $\pi = 3.14$).

[4 Marks]

--- OR ---

From a solid cylinder whose height is 2.4 cm and diameter is 1.4 cm, a conical cavity of the same height and diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm².

37. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60 deg and the angle of depression of its foot is 45 deg. Determine the height of the tower.

[4 Marks]

38. Evaluate the trigonometric value: $(5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ) / (\sin^2 30^\circ + \cos^2 30^\circ)$.

[4 Marks]

--- OR ---

Prove the trigonometric identity: $(\sin \theta - 2 \sin^3 \theta) / (2 \cos^3 \theta - \cos \theta) = \tan \theta$.

39. If a line intersects sides AB and AC of a triangle ABC at D and E respectively and is parallel to BC, prove that $AD / AB = AE / AC$.

[4 Marks]

--- OR ---

A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

40. If the median of the distribution given below is 28.5, find the missing frequencies x and y (Total Frequency = 60):

[4 Marks]

Class Interval	Frequency
0 - 10	5
10 - 20	x
20 - 30	20
30 - 40	15
40 - 50	y
50 - 60	5
Total	60

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